AI & SOCIETY COURSE

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Part 5

AI'S WORLDWIDE IMPACT

Unit 4: The Environmental Cost of AI

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From political power to physical resources

We just said…

- AI companies race for speed and scale
- Scale needs compute, and compute needs real resources
- So they need political power: lobbying, spending, access to government

… So we need to ask:

- What are those resources? Electricity, land, fuel, water
- What does securing these resources cost
- Who is being impacted

Framing from Allen & Weyl (2024), "The Real Dangers of Generative AI", Journal of Democracy

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This unit: goals

- Learn how to think clearly about the environmental impact of AI
- See which resources AI uses, and at which stages
- Understand why the impact could very different, depending on where you are

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Three narratives

AI is ruining the planet and hurting communities

AI is getting more efficient and only uses a small share of global emissions

AI will help us solve the climate problem

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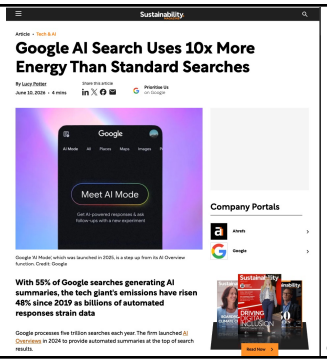
The headlines



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The headlines



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The headlines


Reuters

Tech explores Argentina's Patagonia for mega data centers

By Leila Miller

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What one query costs (2025 estimates)

Google Gemini
0.24 Wh
measured, median Gemini Apps text prompt (May 2025)

OpenAI
0.34 Wh
company figure, average ChatGPT query (model not named)

Epoch AI
0.30 Wh
independent estimate, typical GPT-4o query

Wh = Watt Hours
A 10-watt light bulb left on for 1 hour uses 10 watt-hours of energy. If it's on for 6 minutes, it uses 1 watt-hours.

About twenty seconds of 60w (regular) light bulb.

Google Cloud (2025); Epoch AI (2026)

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But…

1. Those numbers leave out…

- Text prompts only
- Excludes the long tail of heavy queries
- Direct power use; indirect use much larger
- Before "reasoning"
- Larger models = more energy needs
- Unaudited: no transparency!

2. Not all queries cost the same

Text classification **"1" (baseline)**
Text generation **24×**
Image captioning **32×**
Image generation 1,450×

Image generation ≈ 60 × text classification (video is much more).

Luccioni et al. (2024), "Power Hungry Processing"

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!

ALL NUMBERS – EVEN CURRENT NUMBERS -- ARE ESTIMATES BASED ON ASSUMPTIONS!

THERE ARE NO STANDARDS FOR REPORTING AND LITTLE OFFICIAL INFORMATION.

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Queries are a small part of the total

Only 2% of AI electricity is chatbot queries

Where the rest goes:

- Training
- Images, video and audio
- Agents workloads (many calls per task)

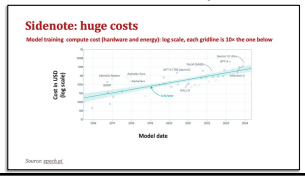
Ritchie, (2026). "How much electricity does AI consume? [2025 summary]"

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The energy needed for training models

- **GPT-3:** about 1.3 GWh, roughly what 120 U.S. homes use in a year
- **GPT-4:** estimated at 50 to 70 GWh, never disclosed. Enough to power a city of 1M people for 1-2 days.
- **Grok 4:** estimated 310 GWh – energy for a town of 20,000 for a two years



Epoch AI (2026); Patterson et al. (2021)

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The energy needed for agent-based queries



OpenAI (2026)

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More than models and queries...

1. Hardware manufacturing
2. Infrastructure construction
3. Dataset creation
4. Model experimentation
5. Training
6. Post-training adaptation
7. Deployment and inference
8. End of life

Lambert & Luccioni (2026), A Life Cycle Review of AI's Environmental Footprint

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The burden by 2030 (estimates)

415 TWh all data centres in 2024, about 1.5% of world electricity

945 TWh all data centres in 2030, close to 3%

IEA, Energy and AI (2025); Ritchie, (2026). "How much electricity does AI consume? [2025 summary]"

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Some measures of energy use

Energy (kWh, in small amounts: Joule)

Carbon emissions, CO_2e

Compute/GPU hours

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Not just energy: different resources...

Land
for the data center, and much more for the power

Fuels
mostly fossil today, and mostly natural gas

Water
for cooling, and for making the electricity

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Land

- The buildings are the small part: most of the land is for generating the power
- For example, 74 gas plants are planned to serve US data centers, about three times California's capacity*



* California is the largest U.S. state in terms of energy

Environmental Integrity Project (2026); UNU-INWEH (2026)

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Fuels

- About 60% of data center electricity today comes from polluting fossil fuels (renewables, nuclear make up the rest)
- The IEA expects gas use for data centers to more than double by 2035



IEA, Energy and AI (2025); Global Energy Monitor, via Carbon Brief (2025); Epoch AI (2026)

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Water

- The water a data center uses on site is almost all cooling: heat from the chips is carried into water that evaporates in cooling towers (leaving the local supply)
- Grok 4 used 750 million liters of cooling water only for training -- drinking water for ~1 million people for a year.
- Add query/inference load, and water used to make the chips and generate the power may be far larger (usually not reported)
- The UN estimates: cost of 9.3 trillion liters water used in 2030; enough for 1.3 billion people water use for one year



Google Cloud (2025); Environmental Integrity Project (2026); Lambert & Luccioni (2026); Epoch.ai

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The same model, different places

- **Ireland:** 23% of all national electricity goes to data centres
- **United States:** about 5% of national electricity

- **The grid mix:** how electricity is produced decides how much carbon a query produces
- **The climate:** decide how much water cooling needs

Where you run a model matters as much as which model you run.

Ireland CSO (2025); IEA; Epoch AI (2026)

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The headlines



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Where we are so far

- Training is huge, but inference/use is a lot more
- Each simple query is small, but workloads can build up
- Resource: energy, land, air, water
- Data centers are heading toward 3% of the world's electricity, and a lot of water…

3% of global electricity by 2030. Is that a lot?

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Is it a lot? One view: AI will pay it back

Better grids, new materials, and AI providing new ideas to fight climate change?

So far:

- A lot of promises, but no promising new directions
- Computation/efficiency improves, but that just enables more use, so no savings.

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Another view: planet on a brink, communities event more

Our planet is already fighting climate change; were headed towards more energy efficiency, now reverse

AI resource use is uneven; communities cannot sustain the resource burden

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Some communities carry more of it

50% of all AI compute capacity sits in five countries

89% of planned US gas plants sit in counties where life expectancy is below average

40% of residents near these plants are low income

Environmental Integrity Project (2026); Oxford (2025)

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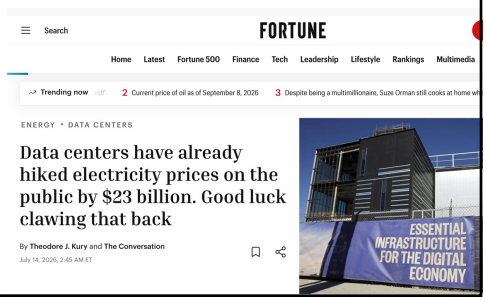
Potential costs for communities

1. Higher energy bills
2. Air and water pollution
3. Water restrictions or limits
4. Noise
5. Strain on the local grid
6. Land use and lost open space
7. Few permanent local jobs
8. Tax breaks shift costs to residents

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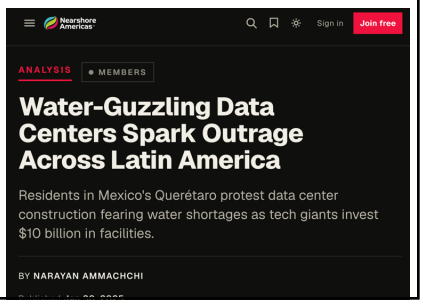
The headlines, again



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The headlines, again



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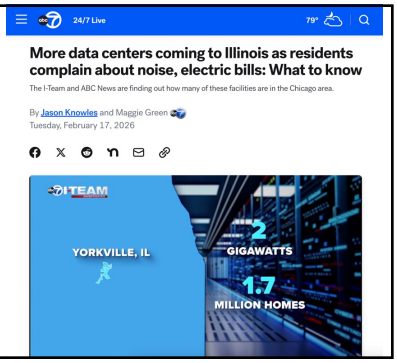
The headlines, again



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The headlines, again



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What the AI companies say

Companies tell local communities that data centers will bring jobs, tax revenue.



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What the AI companies say

A lot of local communities accept this view, some get benefits!

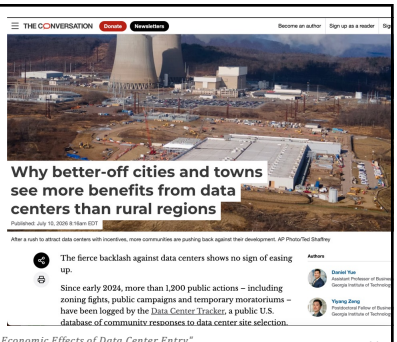
But unclear if the benefits last (for example, construction jobs end after the data center is built).

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Also:

"these benefits are most pronounced when the local economy is more urban and developed."



Source: Yue and Zheng (2026), "The Local Economic Effects of Data Center Entry"

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In my home state



Will Establish a Nation-Leading Regulatory Framework That Protects Ratepayers, the Environment and New Yorkers

One-Year Pause Will Ensure New Yorkers Are Not Paying for Transmission and Infrastructure Build Outs

Will Create New York State Data Center Community Investment Framework, Ensuring Localities Access Can Unlock Millions From New Development

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This unit: summary

- AI may solve climate change, but no evidence yet.
- In the meantime, pushing the boundaries on the planet (pollution, water)…
- … and on local communities, where neighbors get:
  - Pollution
  - Higher electricity bills
  - Noise
  - Water shortages
- Other communities want them for jobs and tax revenue but benefits are unevenly distributed

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What to do

As a user/voter

- Hard to say! Any use of AI drains resources – like driving, make your own judgment.
- Make sure to understand the impact of data center proposals beyond company's promises

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What to do

As a policymaker

- Data centers construction is not necessarily bad
- Better numbers have to come before better policy: require specific, detailed data, reporting all stages, not only training and inference
- Agree on units and always include water and other resources
- Require disclosure and oversight/audting, and tie public funding and contracts to it
- Otherwise you will be taken advantage of – that is what they do.

Lambert & Luccioni (2026), policy implications

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